

ME 333
MECHANICS OF MATERIALS

Lecture 25: Plane Problems in Polar Coordinates

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Plane Problems - Polar Coordinates
Plane Problems in Polar Coordinates

Polar Coordinates : We know that

$$x = r \cos(\theta); \quad y = r \sin(\theta); \quad r = \sqrt{x^2 + y^2}; \quad \theta = \tan^{-1} \left(\frac{y}{x} \right)$$

Considering the differential forms:

$$r_{,x} = \frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2 + y^2}} = \frac{x}{r} = \cos \theta; \quad \theta_{,x} = -\frac{y}{r^2} = -\frac{\sin \theta}{r}$$

$$r_{,y} = \frac{\partial r}{\partial y} = \frac{2y}{2\sqrt{x^2 + y^2}} = \frac{y}{r} = \sin \theta; \quad \theta_{,y} = \frac{x}{r^2} = \frac{\cos \theta}{r}$$

Thus we can write the differentiation formulae as follows:

$$\frac{\partial ()}{\partial x} = \frac{\partial ()}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial ()}{\partial \theta} \frac{\partial \theta}{\partial x} = \frac{\partial ()}{\partial r} \cos \theta - \frac{\partial ()}{\partial \theta} \frac{\sin \theta}{r}$$

$$\frac{\partial ()}{\partial y} = \frac{\partial ()}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial ()}{\partial \theta} \frac{\partial \theta}{\partial y} = \frac{\partial ()}{\partial r} \sin \theta + \frac{\partial ()}{\partial \theta} \frac{\cos \theta}{r}$$

Plane Problems in Polar Coordinates

- From the stress transformation relations we have:

$$\sigma_{rr} = \sigma_{xx} \cos^2 \theta + \sigma_{yy} \sin^2 \theta + 2\sigma_{xy} \sin \theta \cos \theta$$

$$\sigma_{\theta\theta} = \sigma_{xx} \sin^2 \theta + \sigma_{yy} \cos^2 \theta - 2\sigma_{xy} \sin \theta \cos \theta$$

$$\sigma_{r\theta} = (\sigma_{yy} - \sigma_{xx}) \sin \theta \cos \theta + \sigma_{xy} (\cos^2 \theta - \sin^2 \theta)$$

- The stresses relate to the Airy stress function as:

$$\phi_{,xx} = \frac{\partial}{\partial x} \left[\frac{\partial \phi}{\partial x} \right] = \frac{\partial}{\partial x} \left[\frac{\partial \phi}{\partial r} \cos \theta - \frac{\partial \phi}{\partial \theta} \frac{\sin \theta}{r} \right]$$

$$= \phi_{,rr} \cos^2 \theta - 2\phi_{,r\theta} \frac{\sin \theta \cos \theta}{r} + 2\phi_{,\theta\theta} \frac{\sin \theta \cos \theta}{r} + \phi_{,r} \frac{\sin^2 \theta}{r} + \phi_{,\theta\theta} \frac{\sin^2 \theta}{r^2}$$

Similarly, substituting for σ_{yy} and σ_{xy} using $\phi_{,xx}$ and $-\phi_{,xy}$ and collating all the like terms results in:

$$\sigma_{rr} = \frac{1}{r} \phi_{,r} + \frac{1}{r^2} \phi_{,\theta\theta}; \quad \sigma_{\theta\theta} = \phi_{,rr}; \quad \sigma_{r\theta} = \frac{1}{r^2} \phi_{,\theta} - \frac{1}{r} \phi_{,r\theta}$$

Plane Problems in Polar Coordinates

- The sum of the normal stresses can be written as Laplacian of ϕ , i.e.;

$$\nabla^2 \phi = \sigma_{xx} + \sigma_{yy} = \sigma_{rr} + \sigma_{\theta\theta} = \phi_{,rr} + \frac{1}{r} \phi_{,r} + \frac{1}{r^2} \phi_{,\theta\theta}$$

- The biharmonic equation can then be written as :

$$\nabla^4 \phi = \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} \right) \left(\phi_{,rr} + \frac{1}{r} \phi_{,r} + \frac{1}{r^2} \phi_{,\theta\theta} \right) = 0$$

- Note that the biharmonic equation is *invariant*.
- Next step is to recast the stress-strain, strain-displacement and equilibrium relations in terms of polar coordinates.

Plane Problems in Polar Coordinates

- Equilibrium equations:

$$\sigma_{rr,r} + \frac{1}{r}\sigma_{r\theta,\theta} + \frac{1}{r}(\sigma_{rr} - \sigma_{\theta\theta}) = 0$$

$$\sigma_{r\theta,r} + \frac{1}{r}\sigma_{\theta\theta,\theta} + \frac{2}{r}\sigma_{r\theta} = 0$$

- Strain-displacement relations:

$$\epsilon_{rr} = \frac{\partial u_r}{\partial r}; \quad \epsilon_{\theta\theta} = \frac{1}{r}u_{\theta,\theta} + \frac{u_r}{r}; \quad \epsilon_{r\theta} = \frac{1}{2}\left(\frac{1}{r}\frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r}\right)$$

Above relations can also be written as:

$$u_{,r} = \epsilon_{rr}; \quad v_{,\theta} = r\epsilon_{\theta\theta} - u; \quad rv_{,r} + u_{,\theta} = 2r\epsilon_{r\theta} + v$$

Note that $u = u_r$ and $v = u_\theta$ in the above expression.

Plane Problems in Polar Coordinates

- Compatibility relations:

$$\epsilon_{\theta\theta,rr} + \frac{2}{r}\epsilon_{\theta\theta,r} - \frac{1}{r}\epsilon_{rr,r} + \frac{1}{r^2}\epsilon_{rr,\theta\theta} = \frac{2}{r}\epsilon_{r\theta,\theta r} + \frac{2}{r^2}\epsilon_{r\theta,\theta}$$

- Hooke's Law:

$$\epsilon_{rr} = \frac{1}{E}[\sigma_{rr} - \nu\sigma_{\theta\theta}]$$

$$\epsilon_{\theta\theta} = \frac{1}{E}[\sigma_{\theta\theta} - \nu\sigma_{rr}]$$

$$\epsilon_{r\theta} = \frac{1}{2G}\sigma_{r\theta}$$

Note : Since, coordinates are locally orthogonal they have the same form as that of the Cartesian case.

Axisymmetric Problems

Axisymmetric Problems are those problems that exhibit θ independence.

- Airy stress function and stress relations simplify as:

$$\sigma_{rr} = \frac{1}{r}\phi_{,r}; \quad \sigma_{\theta\theta} = \phi_{,rr}; \quad \sigma_{r\theta} = 0$$

- The governing equation in Axisymmetric problems:

$$\nabla^2\phi = \phi_{,rr} + \frac{1}{r}\phi_{,r} = \frac{1}{r}[r(\phi)_{,r}]_{,r}$$

$$\nabla^4\phi = \left[\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{d}{dr}\right]\left[\phi_{,rr} + \frac{1}{r}\phi_{,r}\right] = \frac{1}{r}\left[r\left(\frac{1}{r}[r\phi_{,r}]_{,r}\right)_{,r}\right]_{,r} = 0$$

Note : Above representation is amenable to integration.

Axisymmetric Problems

Solution to the biharmonic equation :

- Integrating the biharmonic equation:

$$\nabla^4\phi = \frac{1}{r}\left[r\left(\frac{1}{r}[r\phi_{,r}]_{,r}\right)_{,r}\right]_{,r} = 0 \implies r\left(\frac{1}{r}[r\phi_{,r}]_{,r}\right)_{,r} = C_1$$

$$\implies \left(\frac{1}{r}[r\phi_{,r}]_{,r}\right)_{,r} = \frac{C_1}{r} \implies \frac{1}{r}[r\phi_{,r}]_{,r} = C_1 \ln r + C_2$$

$$\implies [r\phi_{,r}]_{,r} = C_1 r \ln r + C_2 r \implies r\phi_{,r} = \frac{C_1}{2}r^2 \ln r + \frac{C_2}{2}r^2 + C_3$$

$$\implies \phi_{,r} = \frac{C_1}{2}r \ln r + \frac{C_2}{2}r + \frac{C_3}{r} \implies \phi = \frac{C_1}{4}r^2 \ln r + \frac{C_2}{4}r^2 + C_3 \ln r + C_4$$

Note : The general functional form of ϕ can therefore be written as:

$$\phi = C_1 r^2 \ln r + C_2 r^2 + C_3 \ln r + C_4$$

Thick Walled Cylinders

Solution to the biharmonic equation :

- The stress field can be determined as:

$$\sigma_{rr} = \frac{1}{r} \phi_{,r} = C_1(1 + 2 \ln r) + 2C_2 + \frac{C_3}{r^2}$$

$$\sigma_{\theta\theta} = \phi_{,rr} = C_1(3 + 2 \ln r) + 2C_2 - \frac{C_3}{r^2}$$

$$\sigma_{r\theta} = 0$$

- Checking if the equilibrium conditions are satisfied:

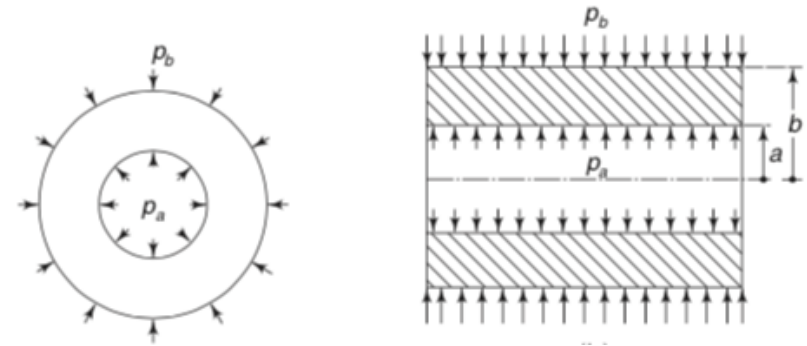
$$\sigma_{rr,r} + \frac{1}{r} (\sigma_{rr} - \sigma_{\theta\theta}) = 0$$

- The strain-displacement relation is:

$$\epsilon_{rr} = \frac{\partial u_r}{\partial r}; \quad \epsilon_{\theta\theta} = \left(\frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r} \right) = \frac{u_r}{r}$$

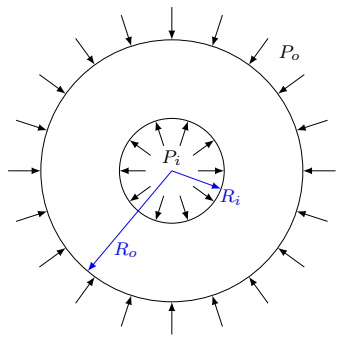
Thick-walled Cylinders

Under the action of internal and external pressures



Ref Base Fig: Advanced Mechanics of Solids, L. S. Srinath

Thick-walled Cylinders



- The radial stress in presence of internal and external pressures is:

$$\sigma_{rr} = -P_o @ r = R_o; \quad \sigma_{rr} = -P_i @ r = R_i$$

- The stresses can therefore be written as:

$$\sigma_{rr} = -P_o = C_1(1 + 2 \ln R_o) + 2C_2 + \frac{C_3}{R_o^2}$$

$$\sigma_{rr} = -P_i = C_1(1 + 2 \ln R_i) + 2C_2 + \frac{C_3}{R_i^2}$$

Problem - two equations in three unknowns

Thick-walled Cylinders

Strain - displacement relations :

- Step 1** : The strain-displacement relations are:

$$\epsilon_{rr} = \frac{\partial u_r}{\partial r}; \quad \epsilon_{\theta\theta} = \left(\frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r} \right) = \frac{u_r}{r}$$

- Step 2** : But from Hooke's Law:

$$\epsilon_{rr} = \frac{1}{E} [\sigma_{rr} - \nu \sigma_{\theta\theta}]; \quad \epsilon_{\theta\theta} = \frac{1}{E} [\sigma_{\theta\theta} - \nu \sigma_{rr}]; \quad \epsilon_{r\theta} = \frac{1}{2G} \sigma_{r\theta}$$

- Step 3** : Also, the stresses in the axi-symmetric case are :

$$\sigma_{rr} = C_1(1 + 2 \ln r) + 2C_2 + \frac{C_3}{r^2}; \quad \sigma_{\theta\theta} = C_1(3 + 2 \ln r) + 2C_2 - \frac{C_3}{r^2}; \quad \sigma_{r\theta} = 0$$

Thick-walled Cylinders

Strain - displacement relations :

- **Step 4** : The displacements obtained by substituting Steps 2 & 3 into 1 and integrating (for the general case not the axisymmetric case) are:

$$u_r = \frac{1}{E} \left[C_1 r [(1 - \nu)(2 \ln r - 1) - 2\nu] + 2C_2(1 - \nu)r - C_3 \frac{(1 + \nu)}{r} + C_4 \sin \theta + C_5 \cos \theta \right]$$

$$u_\theta = \frac{1}{E} [4C_1 r \theta + C_4 \cos \theta - C_5 \sin \theta + C_6 r]$$

- Note that $u_\theta(\theta)$ is multivalued $\therefore u_\theta(0) = u_\theta(2\pi) = u_\theta(4\pi)$ etc.. Therefore, $c_1 = 0$.
- Considering $C_1 = 0$, solving the two equations for stresses simultaneously, we obtain:

$$C_2 = \frac{P_i R_i^2 - P_o R_o^2}{2(R_o^2 - R_i^2)}; \quad C_3 = \frac{R_i^2 R_o^2 (P_o - P_i)}{(R_o^2 - R_i^2)}$$

Thick-walled Cylinders

- Substituting for C_2 and C_3 the stresses take the form:

$$\sigma_{rr} = \frac{P_i R_i^2 - P_o R_o^2}{(R_o^2 - R_i^2)} + \frac{R_i^2 R_o^2 (P_o - P_i)}{r^2 (R_o^2 - R_i^2)}$$

$$\sigma_{\theta\theta} = \frac{P_i R_i^2 - P_o R_o^2}{(R_o^2 - R_i^2)} - \frac{R_i^2 R_o^2 (P_o - P_i)}{r^2 (R_o^2 - R_i^2)}$$

- The stresses in plane do not depend upon elastic constants.
- In case of plane stress $\sigma_{zz} = 0$. Then:

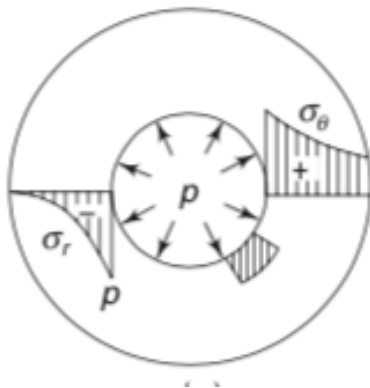
$$\epsilon_{zz} = -\frac{\nu}{E} (\sigma_{rr} + \sigma_{\theta\theta}) = \nu (\sigma_{xx} + \sigma_{yy}) = \frac{2\nu (P_o R_o^2 - P_i R_i^2)}{E (R_o^2 - R_i^2)}$$

- In case of plane strain $\epsilon_{zz} = 0$. Then:

$$\sigma_{zz} = \nu (\sigma_{rr} + \sigma_{\theta\theta}) = \nu (\sigma_{xx} + \sigma_{yy}) = 2\nu \frac{(P_i R_i^2 - P_o R_o^2)}{(R_o^2 - R_i^2)}$$

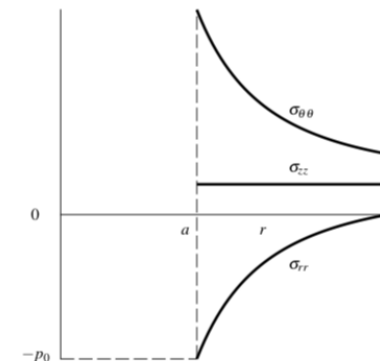
Thick-walled Cylinders

Stress Profiles : In case of internal pressure only



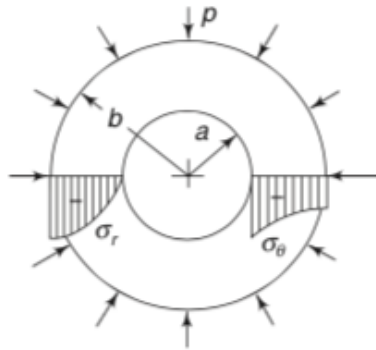
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Stress Profiles : In a general case :



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Stress Profiles : In case of external pressure only



Ref Base Fig: Advanced Mechanics of Solids, L. S. Srinath

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Thick-walled Cylinders

Example : Consider if there is no hole at the origin. Then there exists a material point even at $r = 0$.

- Using general solution :

$$\sigma_{rr} = C_1(1 + 2 \ln r) + 2C_2 + C_3 \frac{1}{r^2}$$

$$\sigma_{\theta\theta} = C_1(3 + 2 \ln r) + 2C_2 + C_3 \frac{1}{r^2}$$

$$\sigma_{r\theta} = 0$$

- In this case, $C_1 = 0$ and $C_3 = 0$. It results in $\sigma_{rr} = \sigma_{\theta\theta} = 2C_2$.
- Assuming no body forces - it is only possible to have a constant state of stress.

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Working Assignment

Derive all relations in Polar coordinates

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